

Differential Calculus — Complete Notes

Grade 12

Mathematics

CAPS

What you'll learn

- What a limit is and how it leads to the derivative.
- How to find the derivative from first principles.
- The power rule and how to differentiate sums, constants and roots.
- How to find the gradient at a point and the equation of a tangent line.
- How to locate and classify stationary points on a cubic function.
- How to sketch cubic graphs and solve practical optimisation problems.

The big picture

Calculus is the mathematics of change. In Grade 12 you study **differential calculus**, which answers one central question: *how fast is something changing at a particular instant?*

Think of a car's speedometer. It does not tell you the average speed for the whole trip — it tells you the speed **at that exact moment**. The number on the speedometer is a derivative. Everything in this chapter is built to find that instantaneous rate of change.

CAPS examiners almost always combine three skills in one question: differentiate correctly, use the derivative to find a gradient or turning point, and then interpret the answer in context. So do not just memorise the rules — practise using them.

Limits — the foundation

A **limit** describes the value a function approaches as the input gets closer and closer to some number — without necessarily reaching it.

We write:

$$\lim_{(h \rightarrow 0)} f(x + h) = L$$

This means "as h gets very small, $f(x + h)$ approaches L ". We never actually set $h = 0$, because that would create division by zero. We simplify first, then let h shrink.

Key idea: if you substitute and get $0/0$, the expression can usually be factorised or simplified to remove the problem.

The derivative from first principles

The **derivative** of a function $f(x)$ is the limit of the average gradient as the interval shrinks to zero:

$$f'(x) = \lim_{(h \rightarrow 0)} [f(x + h) - f(x)] / h$$

This is called **differentiation from first principles**. The formula comes directly from the gradient formula $m = (y_2 - y_1)/(x_2 - x_1)$, where the two points are $(x, f(x))$ and $(x + h, f(x + h))$.

Gradient of secant through $(x, f(x))$ and $(x+h, f(x+h))$

$$m = \frac{\text{rise}}{\text{run}} = \frac{f(x+h) - f(x)}{h}$$

As $h \rightarrow 0$ the secant becomes the TANGENT,
and its gradient becomes $f'(x)$.

Notation

There are several ways to write the derivative. They all mean the same thing:

Notation	Read as	Used when
$f'(x)$	"f prime of x"	function form
dy/dx	"dee y dee x"	y is the function
$D_x[f(x)]$	"D of f of x"	operator form

Rules for differentiation

Once you know the rules, you rarely use first principles again — but you must understand it because it is often examined directly.

The power rule

If $f(x) = x^n$, then $f'(x) = n \cdot x^{n-1}$. Multiply by the old power, then reduce the power by one.

The constant rule

The derivative of a constant is 0. A constant does not change, so its rate of change is zero.

Constant multiple rule

$d/dx [k \cdot f(x)] = k \cdot f'(x)$. A constant multiplier stays outside and is carried through.

Sum and difference rule

Differentiate each term separately: $d/dx [f(x) \pm g(x)] = f'(x) \pm g'(x)$.

Roots and fractions

Rewrite them as powers first:

- $\sqrt{x} = x^{(1/2)}$, so its derivative is $(1/2) \cdot x^{(-1/2)} = 1/(2\sqrt{x})$
- $1/x = x^{-1}$, so its derivative is $-x^{-2} = -1/x^2$

Gradient at a point and the tangent line

The derivative gives the **gradient of the tangent** at any point. To find the equation of a tangent at $x = a$:

1. Find the point: calculate $f(a)$ to get the y-coordinate.

- Find the gradient: calculate $f'(a)$.
- Use the straight-line formula: $y - y_1 = m(x - x_1)$.

Stationary points and the shape of a cubic

A **stationary point** is where the gradient is zero: $f'(x) = 0$. These are the turning points (or flat points) of the graph.

To classify each stationary point, use the second derivative $f''(x)$:

Condition at $x = a$	Nature	Shape
$f'(a) = 0$ and $f''(a) < 0$	local maximum	peak
$f'(a) = 0$ and $f''(a) > 0$	local minimum	valley
$f'(a) = 0$ and $f''(a) = 0$	point of inflection	investigate further

A cubic function $f(x) = ax^3 + bx^2 + cx + d$ can have at most two turning points and always has exactly one point of inflection.

Sketching a cubic function

Follow this checklist every time:

- Find the **y-intercept** by setting $x = 0$.
- Find the **x-intercepts** by factorising $f(x) = 0$.
- Find the **stationary points** from $f'(x) = 0$.
- Classify them using $f''(x)$.
- Plot the points and join with a smooth curve. A positive cubic rises on the right; a negative cubic falls on the right.

```

local max
  (1,5)      Cubic  y = x^3 - 6x^2 + 9x + 1
    *          rises from the bottom-left, peaks
  / \         at (1,5), dips to a valley at (3,1),
 /   \       then rises steeply again.
/       \ local min
/         * (3,1)
/

```

Worked example 1 — first principles

Find the derivative of $f(x) = 3x^2 - 2x$ from first principles.

Step 1 — write $f(x + h)$. $f(x + h) = 3(x + h)^2 - 2(x + h) = 3(x^2 + 2xh + h^2) - 2x - 2h = 3x^2 + 6xh + 3h^2 - 2x - 2h$

Step 2 — subtract $f(x)$. $f(x + h) - f(x) = (3x^2 + 6xh + 3h^2 - 2x - 2h) - (3x^2 - 2x) = 6xh + 3h^2 - 2h$

Step 3 — divide by h . $[f(x + h) - f(x)] / h = 6x + 3h - 2$

Step 4 — let $h \rightarrow 0$. $f'(x) = 6x - 2$

Worked example 2 — different rules

Differentiate $f(x) = 5x^4 - 3\sqrt{x} + 2/x$.

Step 1 — rewrite with powers. $f(x) = 5x^4 - 3x^{1/2} + 2x^{-1}$

Step 2 — apply the power rule to each term. $f'(x) = 20x^3 - 3 \cdot (1/2)x^{-1/2} + 2 \cdot (-1)x^{-2}$

Step 3 — simplify. $f'(x) = 20x^3 - 3/(2\sqrt{x}) - 2/x^2$

Worked example 3 — sketch a cubic

Sketch $f(x) = x^3 - 6x^2 + 9x + 1$.

y-intercept: $f(0) = 1$, so the point is $(0, 1)$.

Stationary points: $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$. So $f'(x) = 0$ when $x = 1$ or $x = 3$. $f(1) = 1 - 6 + 9 + 1 = 5 \rightarrow (1, 5)$ $f(3) = 27 - 54 + 27 + 1 = 1 \rightarrow (3, 1)$

Classify: $f''(x) = 6x - 12$. $f''(1) = -6 < 0 \rightarrow$ local maximum at $(1, 5)$. $f''(3) = 6 > 0 \rightarrow$ local minimum at $(3, 1)$.

Sketch: the curve rises from the bottom-left, reaches a peak at $(1, 5)$, dips to a valley at $(3, 1)$, then rises steeply. It crosses the y-axis at $y = 1$.

Common mistakes

- **Setting $h = 0$ too early.** You must simplify the fraction before letting h approach zero, otherwise you get $0/0$.
- **Dropping the sign when differentiating $1/x$.** The derivative of x^{-1} is $-x^{-2}$, not x^{-2} .
- **Confusing $f(x)$ with $f'(x)$.** $f(a)$ gives a y-value; $f'(a)$ gives a gradient. They are different numbers.
- **Forgetting to find the y-coordinate.** When asked for a turning point, give both coordinates, not just x .
- **Misclassifying turning points.** Always test with $f''(x)$; do not assume the first one is a maximum.
- **Not rewriting roots and fractions as powers** before differentiating — the power rule only works on powers.

Quick summary / cheat sheet

Concept	Formula
First principles	$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$
Power rule	$d/dx (x^n) = n \cdot x^{n-1}$
Constant	$d/dx (k) = 0$
Sum rule	$d/dx (f \pm g) = f' \pm g'$
Tangent gradient	$m = f'(a)$
Tangent line	$y - y_1 = m(x - x_1)$
Stationary points	solve $f'(x) = 0$
Max / min test	$f''(x) < 0$ max · $f''(x) > 0$ min

Quick memory hook: **multiply by the power, subtract one from the power.**

Practice questions

1. Differentiate $f(x) = x^2 + 3x$ from first principles.
2. Differentiate $f(x) = 2x^5 - 8x^2 + 7$.
3. Find the gradient of the tangent to $f(x) = x^3 - 6x$ at $x = -1$.
4. Find and classify the stationary points of $f(x) = x^3 - 3x^2 - 9x + 5$.
5. Find the equation of the tangent to $f(x) = x^2 - 4x + 1$ at $x = 3$.
6. A rectangular enclosure is fenced on three sides using 40 m of fencing (the fourth side is a wall). Find the dimensions that give the maximum area.

Answers

1. $f(x+h) - f(x) = 2xh + h^2 + 3h$; divide by $h \rightarrow 2x + h + 3$; $h \rightarrow 0$ gives **$f'(x) = 2x + 3$** .
2. **$f'(x) = 10x^4 - 16x$** .
3. $f'(x) = 3x^2 - 6$, so $f'(-1) = 3 - 6 = -3$.
4. $f'(x) = 3(x - 3)(x + 1)$: stationary at $x = 3$ and $x = -1$. $f(3) = -22$ and $f(-1) = 10$. $f''(x) = 6x - 6$, so $f''(3) = 12 > 0 \rightarrow$ **local minimum at (3, -22)** and $f''(-1) = -12 < 0 \rightarrow$ **local maximum at (-1, 10)**.
5. $f(3) = -2$ and $f'(3) = 2$, so $y + 2 = 2(x - 3)$, giving **$y = 2x - 8$** .
6. Let the two equal sides be x , so the third side is $40 - 2x$. Area $A = 40x - 2x^2$. $A'(x) = 40 - 4x = 0 \rightarrow x = 10$. The sides are **10 m, 10 m and 20 m**, giving a maximum area of **200 m²** ($A'' = -4 < 0$ confirms a maximum).

Official EduRank Team notes. Always cross-check with your textbook and teacher.